

Memory Interfacing: RAM & ROM Decoding MPCA Module 4

> **Definition:** Memory Interfacing is the electronic hardware arrangement that connects semiconductor memory chips (RAM and EPROM/ROM) to the microprocessor's Address, Data, and Control buses using **Address Decoding Logic** to assign unique physical address ranges to each memory chip.

1. Detailed Technical Explanation

1. 8086 Memory Organization Review

- The 8086 microprocessor has a **20-bit Address Bus** (**A₀ - A₁₉**), enabling it to address up to **1 MB** (2^{20} bytes) of physical memory.

- Phy

Microp

2. Address Decoding Logic (74LS138 3-to-8 Decoder)

The **74LS138** decoder is the industry standard IC used to generate chip select (**$\overline{CS}$**) signals for multiple memory chips.

74LS1

Memory Map Design Example:

To interface a 64 KB EPROM chip (Address lines **A₀ - A₁₅** used internally inside chip):

- Hig

3. Core Concepts & Memory Keywords

- **Address Decoding:** Translating upper address bus bits into unique **$\overline{CS}$** chip select signals.

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4. Must-Write Points for Exams

- ROM/EPROM is always placed at the top of memory (`FFFF0H`) because 8086 starts execution at `FFFF0H` upon reset.

- RA

5. Quick Recall Flow

20-bit